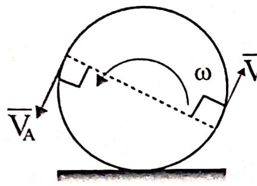
Chapter 1 Engineering Mechanics

One Mark Questions

01. A and B are the end-points of a diameter of a disc rolling along a straight line with a counter Clock-wise angular velocity as shown in the figure. Referring to the velocity vectors $\overline{V}_A$ and $\overline{V}_B$ shown in the figure: **(GATE-ME-90)**

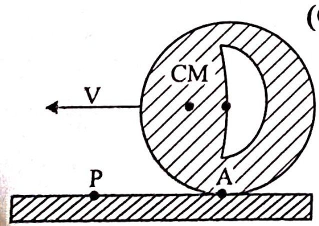


(a) $\overline{V}_A$ and $\overline{V}_B$ are both correct.
(b) $\overline{V}_A$ is incorrect but $\overline{V}_B$ is correct.
(c) $\overline{V}_A$ and $\overline{V}_B$ are both incorrect.
(d) $\overline{V}_A$ is correct but $\overline{V}_B$ is incorrect.

02. Instantaneous centre of a body rolling with sliding on a stationary curved surface lies **(GATE-ME-92)**
(a) at the point of contact
(b) on the common normal at the point of contact
(c) on the common tangent at the point of contact
(d) at the centre of curvature of the stationary surface

03. The cylinder shown below rolls without slipping. Towards which of the following points is the acceleration of the point of contact A on the cylinder directed?

(GATE-ME-93)



(a) The mass centre
(b) The geometric centre
(c) The point P as marked
(d) None of the above

04. A stone of mass m at the end of a string of length ℓ is whirled in a vertical circle at a constant speed. The tension in the string will be maximum when the stone is **(GATE-ME-94)**
(a) at the top of the circle
(b) half way down from the top
(c) quarter-way down from the top
(d) at the bottom of the circle

05. A ball A of mass m falls under gravity from a height h and strikes another ball B of mass m which is supported at rest on a spring of stiffness k. Assume perfect elastic impact. Immediately after the impact **(GATE-ME-96)**
(a) the velocity of ball A is $\frac{1}{2}\sqrt{2gh}$
(b) the velocity of ball A is zero
(c) the velocity of both balls is $\frac{1}{2}\sqrt{2gh}$
(d) none of the above

06. A wheel of mass m and radius r is in accelerated rolling motion without slip under a steady axle torque T. If the coefficient of kinetic friction is μ, the friction force from the ground on the wheel is **(GATE-ME-96)**

(a) μmg (b) T/r
(c) zero (d) none of the above

07. A car moving with uniform acceleration covers 450 m in a 5 second interval, and covers 700 m in the next 5 second interval. The acceleration of the car is **(GATE-ME-98)**
(a) 7 m/s^2 (b) 50 m/s^2 (c) 25 m/s^2 (d) 10 m/s^2

08. A steel wheel of 600 mm diameter is on a horizontal steel rail. It carries a load of 500 N. The coefficient of rolling resistance is 0.3. The force in Newton, necessary to roll the wheel along the rail is **(GATE-ME-00)**
(a) 0.5 (b) 5 (c) 1.5 (d) 150

09. The ratio of tension on the tight side to that on the slack side in a flat belt drive is **(GATE-ME-00)**

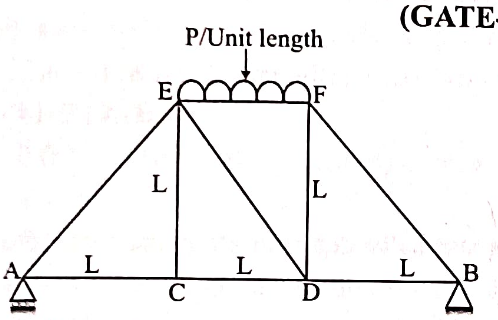
(a) proportional to the product of coefficient of friction and lap angle.

(b) an exponential function of the product of coefficient of friction and lap angle.

(c) proportional to lap angle

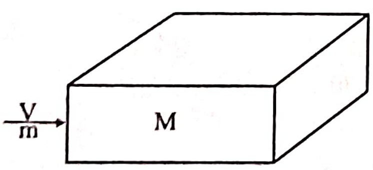
(d) proportional to the coefficient of friction.

10. A truss consists of horizontal members and vertical members having length L each. The members AE, DE and BF are inclined at 45° to the horizontal. For the uniformly distributed load P per unit length on the member EF of the truss shown in the figure, the force in the member CD is **(GATE-ME-03)**



(a) $\frac{PL}{2}$ (b) PL (c) Zero (d) $\frac{2PL}{3}$

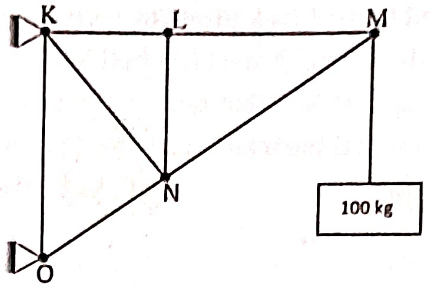
11. A bullet of mass 'm' travels at a very high velocity 'V' (as shown in the figure) and gets embedded inside the block of mass 'M' initially at rest on a rough horizontal floor. The block with the bullet is seen to move a distance 'S' along the floor. Assuming μ to be the coefficient of kinetic friction between the block and the floor and 'g' the acceleration due to gravity, what is the velocity 'V' of the bullet? **(GATE-ME-03)**



(a) $\frac{M+m}{m}\sqrt{2\mu gS}$ (b) $\frac{M-m}{m}\sqrt{2\mu gS}$

(c) $\frac{\mu(M+m)}{m}\sqrt{2gS}$ (d) $\frac{M}{m}\sqrt{2\mu gS}$

12. The figure shows a pin-jointed plane truss loaded at the point M by hanging a mass of 100 kg. The member LN of the truss is subjected to a load of **(GATE-ME-04)**



(a) Zero (b) 490 N in compression

(c) 981 N in compression (d) 981 N in tension

13. The time variation of the position of a particle in rectilinear motion is given by $x = 2t^3 + t^2 + 2t$. If 'V' is the velocity and 'a' acceleration of the particle in consistent units, the motion started with **(GATE-ME-05)**

(a) V = 0, a = 0 (b) V = 0, a = 2

(c) V = 2, a = 0 (d) V = 2, a = 2

14. A simple pendulum of length of 5m, with a bob of mass 1kg, is in simple harmonic motion. As it passes through its mean position, the bob has a speed of 5m/s. The net force on the bob at the mean position is **(GATE-ME-05)**

(a) zero (b) 2.5 N (c) 5 N (d) 25 N

15. During inelastic collision of two particles, which one of the following is conserved? **(GATE-ME-07)**

(a) Total linear momentum only

(b) Total kinetic energy only

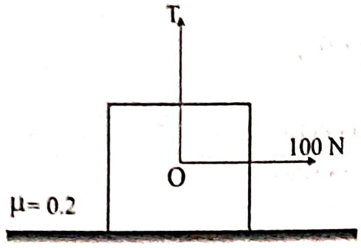
(c) Both linear momentum and kinetic energy

(d) Neither linear momentum nor kinetic energy

16. A straight rod length L(t), hinged at one end freely extensible at the other end, rotates through an angle θ(t) about the hinge. At time t, L(t) = 1m, $\dot{L}(t) = 1$ m/s, θ(t) = π/4 rad and $\dot{\theta}(t) = 1$ rad/s. The magnitude of the velocity at the other end of the rod is **(GATE-ME-08)**

(a) 1 m/s (b) $\sqrt{2}$ m/s (c) $\sqrt{3}$ m/s (d) 2 m/s

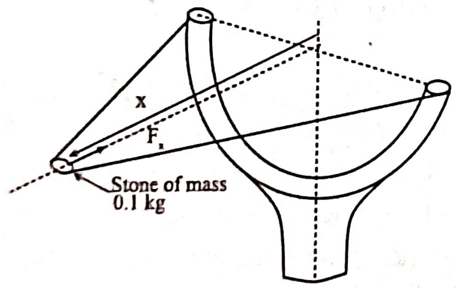
17. A block weighing 981 N is resting on a horizontal surface. The coefficient of friction between the block and the horizontal surface is $\mu = 0.2$. A vertical cable attached to the block provides partial support as shown in the figure. A man can pull horizontally with a force of 100 N. What will be the tension, T (in N) in the cable if the man is just able to move the block to the right? (GATE-ME-09)



(a) 176.2 (b) 196.0 (c) 481.0 (d) 981.0

18. The coefficient of restitution of a perfectly plastic impact is (GATE-ME-11)
(a) Zero (b) 1 (c) 2 (d) infinite

19. A stone with mass of 0.1kg is catapulted as shown in the figure. The total force F_x (in N) exerted by the rubber band as a function of distance x (in m) is given by $F_x = 300x^2$. If the stone is displaced by 0.1 m from the un-stretched position (x = 0) of the rubber band, the energy stored in the rubber band is

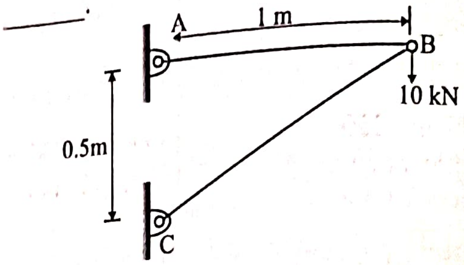


(GATE-ME-11)

(a) 0.01 J (b) 0.1 J (c) 1 J (d) 10 J

20. A circular object of radius r rolls without slipping on a horizontal level floor with the center having velocity V. The velocity at the point of contact between the object and the floor is (GATE-14-SET-1)
(a) zero
(b) V in the direction of motion
(c) V opposite to the direction of motion
(d) V vertically upward from the floor

21. A two member truss ABC is shown in the figure. The force (in kN) transmitted in member AB is ______. (GATE-14-SET-2)

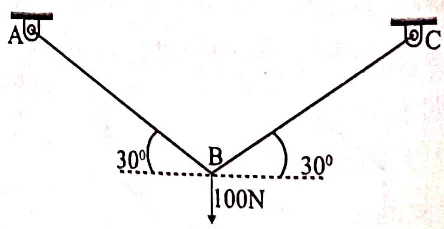


22. A mass m_1 of 100kg traveling with a uniform velocity of 5 m/s along a line collides with a stationary mass m_2 of 1000kg. After the collision, both the masses travel together with the same velocity. The coefficient of restitution is (GATE-14 - SET-3)

(a) 0.6 (b) 0.1 (c) 0.01 (d) 0

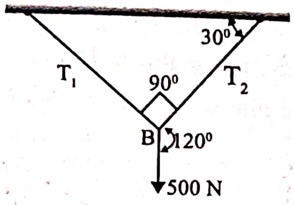
23. In a statically determinate plane truss, the number of joints (j) and the number of members (m) are related by (GATE-14 - SET-4)
(a) $j = 2m - 3$ (b) $m = 2j + 1$
(c) $m = 2j - 3$ (d) $m = 2j - 1$

24. Two identical trusses support a load of 100 N as shown in the figure. The length of each truss is 1.0 m; cross-sectional area is 200 mm², Young's modulus E = 200 GPa. The force in the truss AB (in N) is ______ (GATE –15 – Set 1)



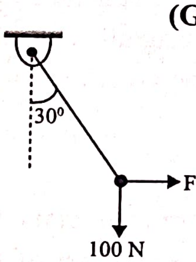
25. A small ball of mass 1 kg moving with a velocity of 12 m/s undergoes a direct central impact with a stationary ball of mass 2 kg. The impact is perfectly elastic. The speed (in m/s) of 2 kg mass ball after the impact will be______ (GATE –15 –Set 2)

26. A weight of 500 N is supported by two metallic ropes as shown in the figure. The values of tensions T_1 and T_2 are respectively. **(GATE –15 –Set 3)**

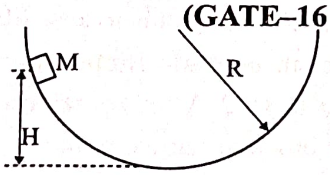


(a) 433 N and 250 N
(b) 250 N and 433 N
(c) 353.5 N and 250 N
(d) 250 N and 353.5 N

27. A rigid ball of weight 100 N is suspended with the help of a string. The ball is pulled by a horizontal force F such that the string makes an angle of 30° with the vertical. The magnitude of force F (in N) is ________. **(GATE – 16 – SET – 1)**

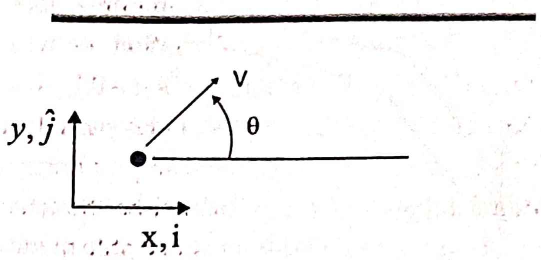


28. A point mass M is released from rest and slides down a spherical bowl (of radius R) from a height H as shown in the figure below. The surface of the bowl is smooth (no friction). The velocity of the mass at the bottom of the bowl is **(GATE–16 – SET – 1)**



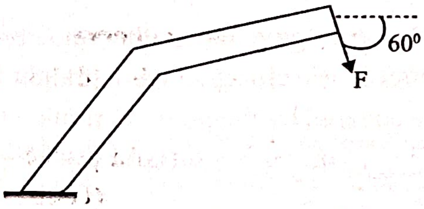
(a) $\sqrt{gH}$ (b) $\sqrt{2gR}$ (c) $\sqrt{2gH}$ (d) 0

29. A point mass having mass M is moving with a velocity V at an angle θ to the wall as shown in the figure. The mass undergoes a perfectly elastic collision with the smooth wall and rebounds. The total change (final minus initial) in the momentum of the mass is **(GATE – 16 – SET – 2)**



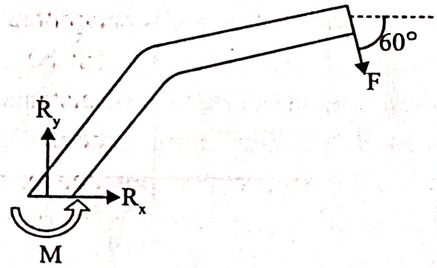
(a) $-2MV\cos\theta\ \hat{j}$
(b) $2MV\sin\theta\ \hat{j}$
(c) $2MV\cos\theta\ \hat{j}$
(d) $-2MV\sin\theta\ \hat{j}$

30. A force F is acting on a bent bar which is clamped at one end as shown in the figure.

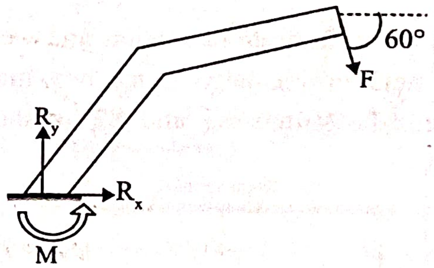


The CORRECT free body diagram is **(GATE – 16 – SET – 3)**

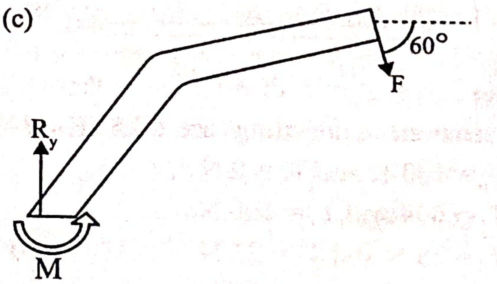
(a)



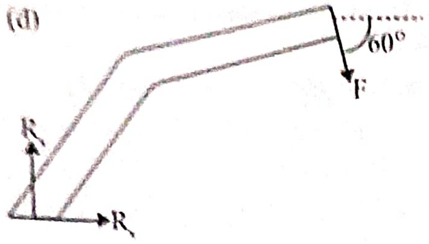
(b)



(c)

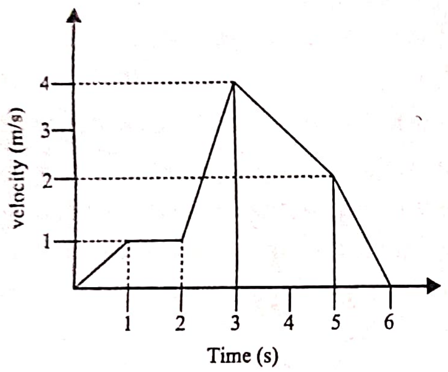


(d)

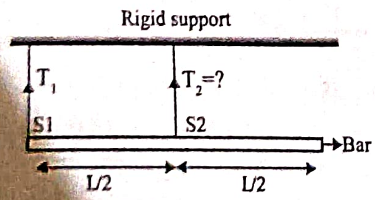


31. A Particle of unit mass is moving on a plane. Its trajectory, in polar coordinates, is given by $r(t) = t^2$, $\theta(t) = t$, where t is time. The kinetic energy of the particle at time t = 2 is **(GATE – 17 – SET – 1)**
(a) 4 (b) 12 (c) 16 (d) 24

32. The following figure shows the velocity-time plot for a particle travelling along a straight line. The distance covered by the particle from t = 0 to t = 5 s is ________ m. **(GATE – 17 – SET – 1)**

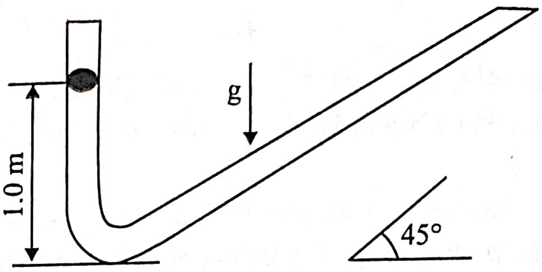


33. A bar of uniform cross section and weighing 100 N is held horizontally using two massless and inextensible strings S1 and S2 as shown in the figure.

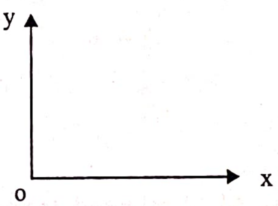


The tensions in the strings are **(GATE–18–SET–1)**
(a) T_1 = 100 N and T_2 = 0 N
(b) T_1 = 0 N and T_2 = 100 N
(c) T_1 = 75 N and T_2 = 25 N
(d) T_1 = 25 N and T_2 = 75 N

34. A ball is dropped from rest from a height of 1 m in a frictionless tube as shown in the figure. If the tube profile is approximated by two straight lines (ignoring the curved portion), the total distance travelled (in m) by the ball is ______ (correct to two decimal places). **(GATE – 18 – SET – 2)**



35. Considering the coordinate system shown in the figure, a force of magnitude 10 kN has x-component of –6 kN. Possible y-component(s) of the force is / are **(PI_GATE – 18)**



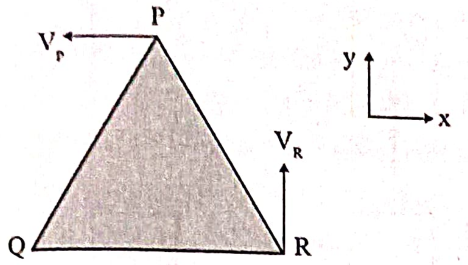
(a) +8 kN only (b) +5 kN only
(c) +8 kN and –8 kN (d) +5 kN and –5 kN

36. A block of mass 10 kg rests on a horizontal floor. The acceleration due to gravity is 9.81 m/s². The coefficient of static friction between the floor and the block is 0.2. A horizontal force of 10 N is applied on the block as shown in the figure. The magnitude of force of friction (in N) on the block is ____.

(GATE – 19 – SET – 1)

37. A rigid triangular body PQR, with sides of equal length of 1 unit moves on a flat plane. At the instant shown, edge QR is parallel to the x-axis, and the body moves such that velocities of points P and R are V_P and V_R, in the x and y directions, respectively.

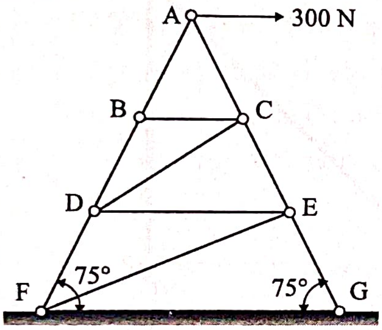
The magnitude of the angular velocity of the body is



(GATE – 19 – SET – 2)

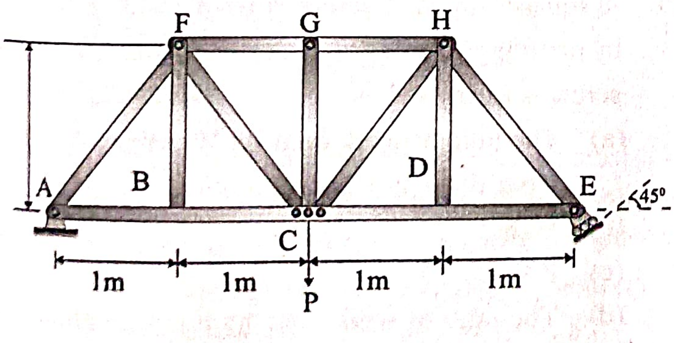
(a) $\frac{V_P}{\sqrt{3}}$ (b) $2V_P$ (c) $\frac{V_R}{\sqrt{3}}$ (d) $2V_R$

38. The figure shows an idealized plane truss. If a horizontal force of 300 N is applied at point A, then the magnitude of the force produced in member CD is ______ N.



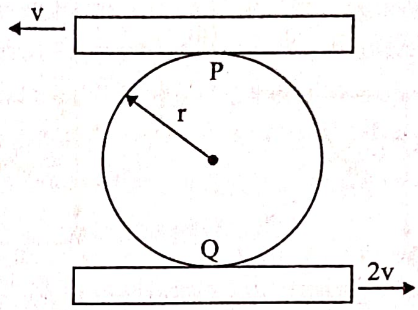
(GATE – 19 – SET – 2)

39. The members carrying zero force (i.e., zero-force members) in the truss shown in the figure, for any load P > 0 with no appreciable deformation of the truss (i.e., with no appreciable change in angles between the members), are **(GATE–20–SET–1)**



(a) BF, DH, GC, CD and DE only
(b) BF and DH only
(c) BF, DH, GC, FG and GH only
(d) BF, DH and GC only

40. A circular disk of radius r is confined to roll without slipping at P and Q as shown in the figure.

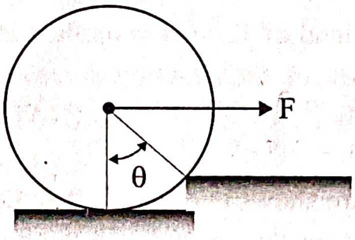


If the plates have velocities as shown, the magnitude of the angular velocity of the disk is

(GATE–20–SET–2)

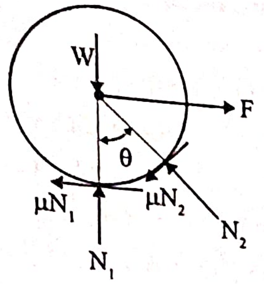
(a) $\frac{v}{r}$ (b) $\frac{v}{2r}$ (c) $\frac{2v}{3r}$ (d) $\frac{3v}{2r}$

41. An attempt is made to pull a roller of weight W over a curb (step) by applying a horizontal force F as shown in the figure.

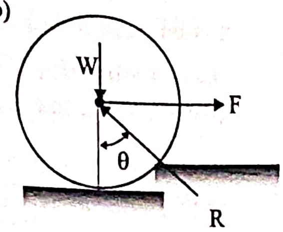


The coefficient of static friction between the roller and the ground (including the edge of the step) is μ. Identify the correct free body diagram (FBD) of the roller when the roller is just about to climb over the step. **(GATE–20–SET–2)**

(a)



(b)



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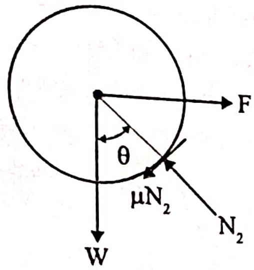
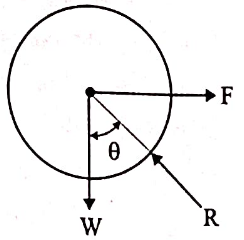
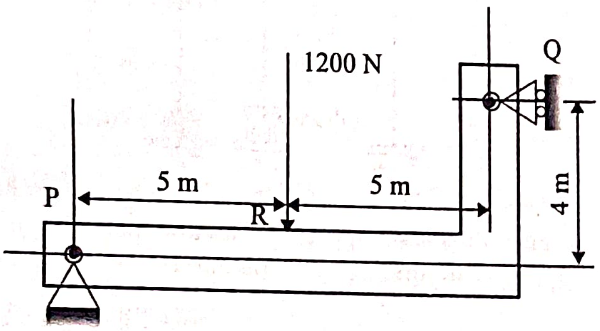


(d)

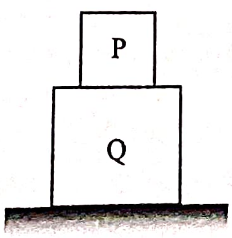


42. A beam of negligible mass in hinged at support P and has a roller support Q as shown in the figure.



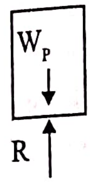
A point load of 1200 N is applied at point R. The magnitude of the reaction force at support Q is ______ N. **(GATE–20–SET–2)**

43. The figure shows two bodies P and Q. The body Q is placed on the ground and the body P is placed on top of it. The weights of P and Q are W_P and W_Q, respectively. The bodies are at rest and all the surfaces are assumed to be frictionless. R represents reaction force, if any, between the bodies.

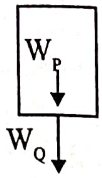


The correct free body diagram of the body P is **(GATE-PI-20)**

(a)



(b)



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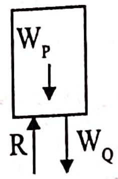

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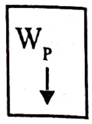
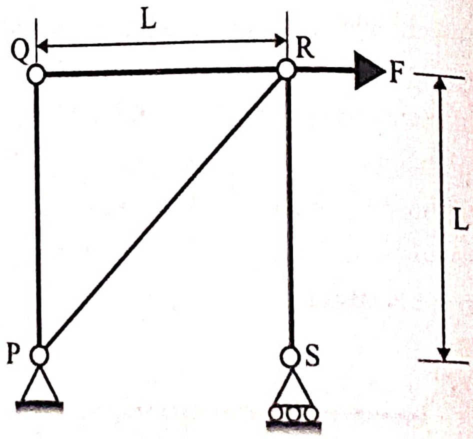


44. A plane truss PQRS (PQ = RS, and $\angle PQR = 90°$) is shown in the figure.



The forces in the members PR and RS, respectively, are **(GATE–21_SET–2)**

(a) F (tensile) and $F\sqrt{2}$ (tensile)

(b) F (compressive) and $F\sqrt{2}$ (compressive)

(c) $F\sqrt{2}$ (tensile) and F (tensile)

(d) $F\sqrt{2}$ (tensile) and F (compressive)

45. A square threaded screw is used to lift a load W by applying a force F. Efficiency of square threaded screw is expressed as **(GATE-22_SET-1)**

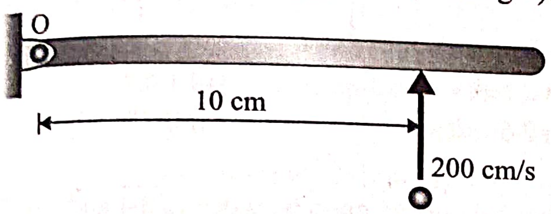
(a) The ratio of work done by W per revolution to work done by F per revolution

(b) W/F

(c) F/W

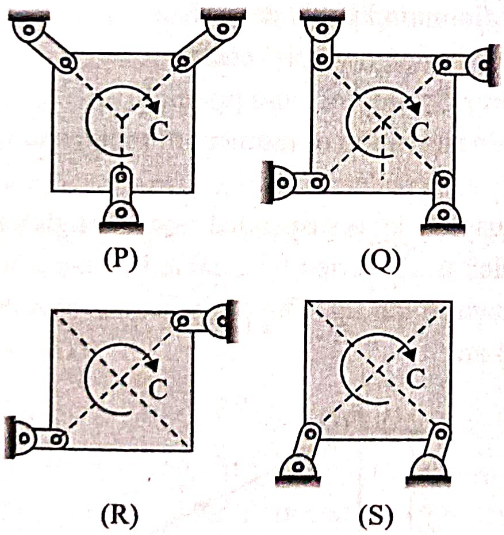
(d) The ratio of work done by F per revolution to work done by W per revolution

46. The plane of the figure represents a horizontal plane. A thin rigid rod at rest is pivoted without friction about a fixed vertical axis passing through O. Its mass moment of inertia is equal to 0.1 kg·cm² about O. A point mass of 0.001 kg hits it normally at 200 cm/s at the location shown, and sticks to it. Immediately after the impact, the angular velocity of the rod is ___________ rad/s (in integer).



(GATE-22_SET-1)

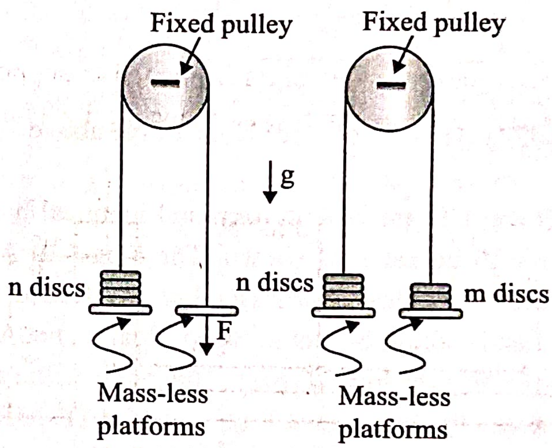
47. A square plate is supported in four different ways (configurations (P) to (S) as shown in the figure). A couple moment C is applied on the plate. Assume all the members to be rigid and mass-less, and all joints to be frictionless. All support links of the plate are identical.



The square plate can remain in equilibrium in its initial state for which one or more of the following support configurations? (GATE-22_SET-2)

(a) Configuration (P) (b) Configuration (Q)
(c) Configuration (R) (d) Configuration (S)

48. A rope with two mass-less platforms at its two ends passes over a fixed pulley as shown in the figure. Discs with narrow slots and having equal weight of 20 N each can be placed on the platforms. The number of discs placed on the left side platform is n and that on the right side platform is m.
It is found that for n = 5 and m = 0, a force F = 200 N (refer to part (i) of the figure) is just sufficient to initiate upward motion of the left side platform. If the force F is removed then the minimum value of m (refer to part (ii) of the figure) required to prevent downward motion of the left side platform is ________ (in integer).



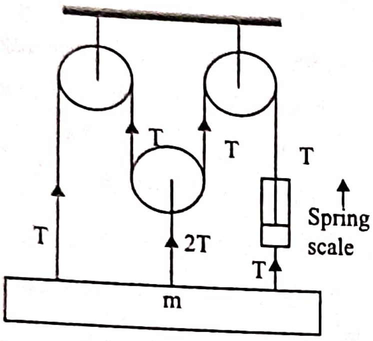
(GATE-22_SET-2)

49. A solid circular disk of 0.025 m thickness is used as flywheel. The density of the disk material is 7800 kg/m³ and the mass moment of inertia of the disk about its center is 4.36 kg-m². The radius, in m, of the disk is ______(round off to 2 decimal places).

(GATE_PI-23)

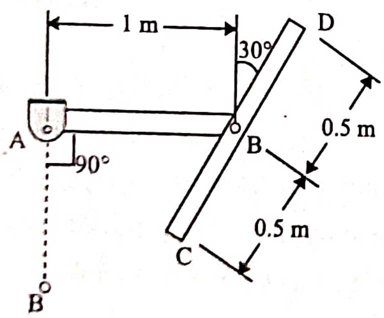
Two Marks Questions

01. A spring scale indicates a tension T in the right hand cable of the pulley system shown in the fig. Neglecting the mass of the pulleys and ignoring friction between the cable and pulley, the mass m is (GATE-ME-95)



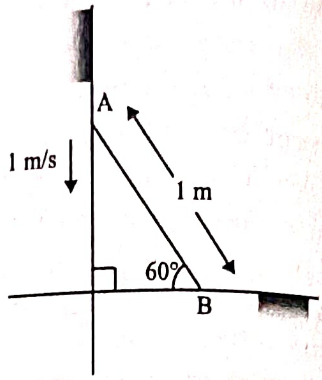
(a) 2T/g
(b) $T(1 + e^{4\pi})/g$
(c) 4T/g
(d) None of the above

02. AB and CD are two uniform and identical bars of mass 10 kg each, as shown. The hinges at A and B are frictionless. The assembly is released from rest and motion occurs in the vertical plane. At the instant that the hinge B passes the point B, the angle between the two bars will be (GATE-ME-96)



(a) 60 degrees
(b) 37.4 degrees
(c) 30 degrees
(d) 45 degrees

03. A rod of length 1m is sliding in a corner as shown in the fig. At an instant when the rod makes an angle of 60 degrees with the horizontal plane, the velocity of point A on the rod is 1m/s. The angular velocity of the rod at this instant is (GATE-ME-96)



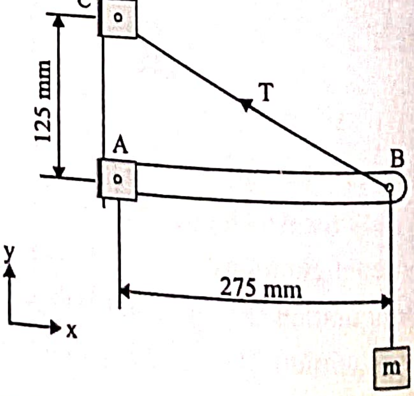
(a) 2 rad/s
(b) 1.5 rad/s
(c) 0.5 rad/s
(d) 0.75 rad/s

04. Match 4 correct pairs between List-I and List-II. No credit will be given for partially correct matching (GATE-ME-96)

List – I
a. Collision of particles
b. Stability
c. Satellite motion
d. Spinning top

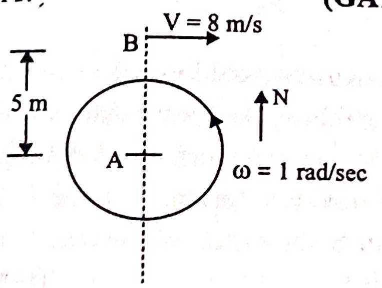
List – II
1. Euler's equation of motion
2. Minimum kinetic energy
3. Minimum potential energy
4. Impulse-momentum principle
5. Conservation of moment of momentum

05. A mass 35 kg is suspended from a weightless bar AB which is supported by a cable CB and a pin at A as shown in the fig. The pin reactions at A on the bar AB are (GATE-ME-97)



(a) $R_x = 343.4$ N, $R_y = 755.4$ N
(b) $R_x = 343.4$ N, $R_y = 0$
(c) $R_x = 755.4$ N, $R_y = 343.4$ N
(d) $R_x = 755.4$ N, $R_y = 0$

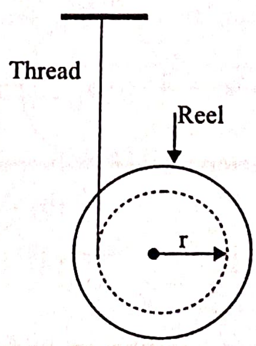
06. As shown in Figure, a person A is standing at the centre of a rotating platform facing person B who is riding a bicycle, heading East. The relevant speeds and distances are shown in given figure person, a bicycle, heading East. At the instant under consideration, what is the apparent velocity of B as seen by A?, **(GATE-ME-99)**



(a) 3 m/s heading East (b) 3 m/s heading West
(c) 8 m/s heading East (d) 13 m/s heading East

Common data Q. 07 & 08

A reel of mass "m" and radius of gyration "k" is rolling down smoothly from rest with one end of the thread wound on it held in the ceiling as depicted in the figure below. Considering the thickness of the thread and its mass negligible in comparison with the radius "r" of the hub and the reel mass "m", symbol "g" represents the acceleration due to gravity. **(GATE-2003)**



07. The linear acceleration of the reel is:

(a) $\dfrac{gr^2}{(r^2+k^2)}$ (b) $\dfrac{gk^2}{(r^2+k^2)}$

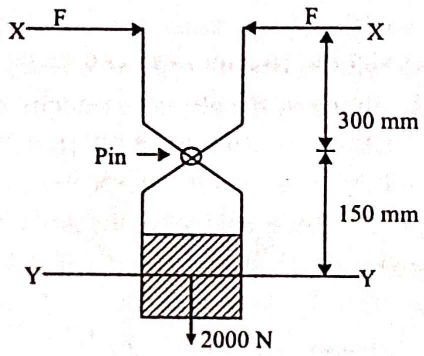
(c) $\dfrac{grk}{(r^2+k^2)}$ (d) $\dfrac{mgr^2}{(r^2+k^2)}$

08. The tension in thread is

(a) $\dfrac{mgr^2}{(r^2+k^2)}$ (b) $\dfrac{mgk^2}{(r^2+k^2)}$

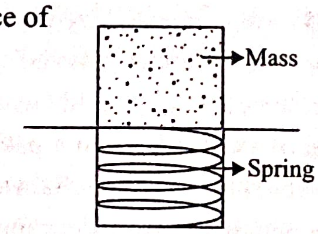
(c) $\dfrac{mgrk}{(r^2+k^2)}$ (d) $\dfrac{mg}{(r^2+k^2)}$

09. The figure below shows a pair of pin jointed gripper tongs holding an object weighting 2000N. The coefficient of friction (μ) at the gripping surface is 0.1. XX is the line of action of the input force and Y – Y is the line of application of gripping force. If the pin joint is assumed to be frictionless, the magnitude of force F required to hold the weight is **(GATE-ME-04)**



(a) 1000 N (b) 2000 N (c) 2500 N (d) 5000 N

10. An ejector mechanism consists of a helical compression spring having a spring constant of k = 981x10³ N/m. It is pre-compressed by 100 mm from its free state. If it is used to eject a mass of 100 kg held on it, the mass will move up through a distance of **(GATE-ME-04)**



(a) 100 mm (b) 5000 mm
(c) 581 mm (d) 1000 mm

11. A rigid body shown in the fig.(a) has a mass of 10kg. It rotates with a uniform angular velocity 'ω' A balancing mass of 20 kg is attached as shown in fig.(b). The percentage increase in mass moment of inertia as a result of this addition is

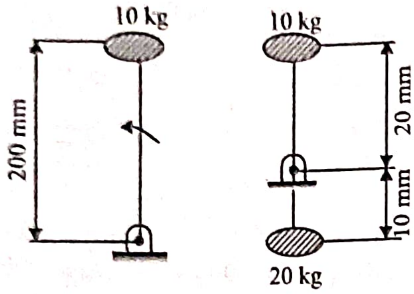
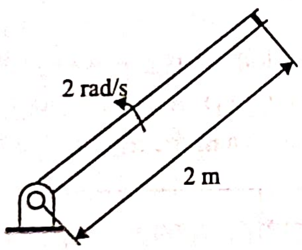


Fig. (a) Fig. (b)

(GATE-ME-04)

(a) 25% (b) 50% (c) 100% (d) 200%

12. A shell is fired from cannon. At the instant the shell is just about to leave the barrel, its velocity relative to the barrel is 3m/s, while the barrel is swinging upwards with a constant angular velocity of 2 rad/s. The magnitude of the absolute velocity of the shell is (GATE-ME-05)



(a) 3 m/s (b) 4 m/s
(c) 5 m/s (d) 7 m/s

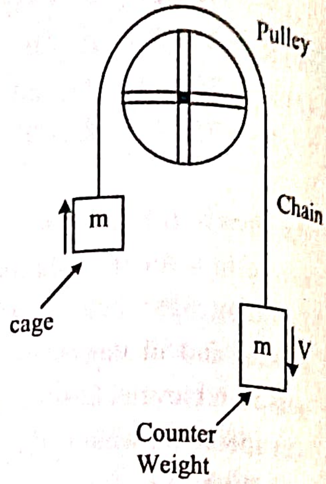
13. An elevator (lift) consists of the elevator cage and a counter weight, of mass m each. The cage and the counter weight are connected by chain that passes over a pulley. The pulley is coupled to a motor. It is desired that the elevator should have a maximum stopping time of t seconds from a peak speed V. If the inertias of the pulley and the chain are neglected, the minimum power that the motor must have is (GATE-ME-05)

(a) $\frac{1}{2}mV^2$

(b) $\frac{mV^2}{2t}$

(c) $\frac{mV^2}{t}$

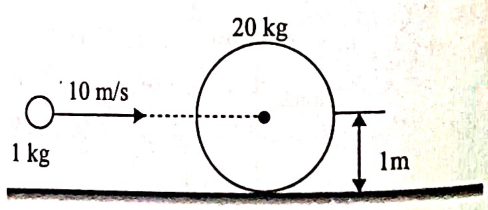
(d) $\frac{2mV^2}{t}$



14. Two books of mass 1 kg each are kept on a table one over the other. The coefficient of friction on every pair of containing surfaces is 0.3. The lower book is pulled with a horizontal force F. The minimum value of F for which slip occurs between the two books is (GATE-ME-05)

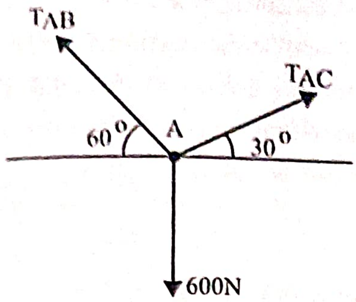
(a) Zero (b) 11.77 N
(c) 5.74 N (d) 8.83 N

15. A 1 kg mass of clay, moving with a velocity of 10 m/s, strikes a stationary wheel and sticks to it. The solid wheel has a mass of 20 kg and a radius of 1 m. Assuming that the wheel and the ground are both rigid and that the wheel is set into pure rolling motion, the angular velocity of the wheel immediately after the impact is approximately (GATE-ME-05)



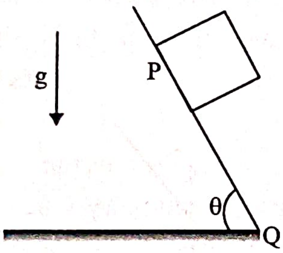
(a) Zero (b) 1/3 rad/s
(c) $\sqrt{\frac{10}{3}}$ rad/s (d) 10/3 rad/s

16. If point A is in equilibrium under the action of the applied forces, the values of tensions T_{AB} and T_{AC} are respectively (GATE-ME-06)



(a) 520 N and 300 N
(b) 300 N and 520 N
(c) 450 N and 150 N
(d) 150 N and 450 N

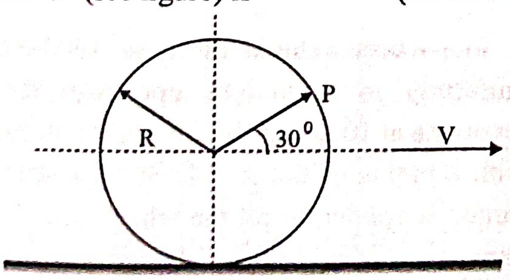
17. A block of mass M is released from point P on a rough inclined plane with inclination angle θ, shown in the figure below. The coefficient of friction is μ. If $\mu < \tan\theta$, then the time taken by the block to reach another point Q on the inclined plane, where PQ = S, is **(GATE-ME-07)**



(a) $\sqrt{\dfrac{2s}{g\cos\theta(\tan\theta-\mu)}}$ (b) $\sqrt{\dfrac{2s}{g\cos\theta(\tan\theta+\mu)}}$

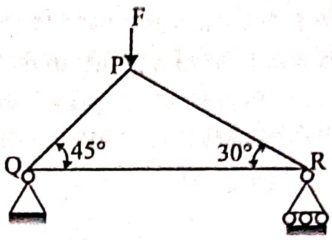
(c) $\sqrt{\dfrac{2s}{g\sin\theta(\tan\theta-\mu)}}$ (d) $\sqrt{\dfrac{2s}{g\sin\theta(\tan\theta+\mu)}}$

18. A circular disk of radius 'R' rolls without slipping at a velocity V. The magnitude of the velocity at point 'P' (see figure) is **(GATE-ME-08)**



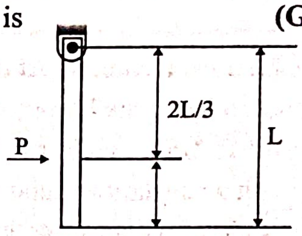
(a) $\sqrt{3}\,V$ (b) $\dfrac{\sqrt{3}\,V}{2}$ (c) $\dfrac{V}{2}$ (d) $\dfrac{2V}{\sqrt{3}}$

19. Consider a truss PQR loaded at P with a force F as shown in the figure. The tension in the member QR is **(GATE- ME-08)**



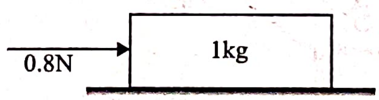
(a) 0.5 F (b) 0.63 F (c) 0.73 F (d) 0.87 F

20. A uniform rigid rod of mass M and length L is hinged at one end as shown in the below figure. A force P is applied at a distance of 2L/3 from the hinge so that the rod swings to the right. The horizontal reaction at the hinge is **(GATE-ME-09)**



(a) –P (b) 0 (c) P/3 (d) 2P/3

21. A 1 kg block is resting on a surface with coefficient of friction μ = 0.1. A force of 0.8 N is applied to the block as shown in the figure. The friction force is

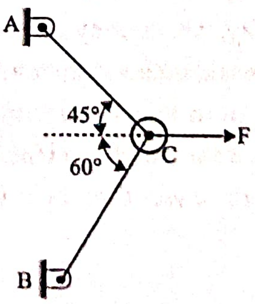


(GATE-ME-11)

(a) Zero (b) 0.8 N (c) 0.89 N (d) 1.2 N

Common Data for Q.Nos. Q.22 & 23

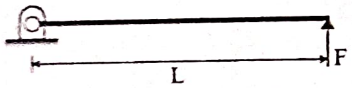
Two steel truss members, AC and BC, each having cross sectional area of 100 mm², are subjected to a horizontal force F as shown in figure. All the joints are hinged. **(GATE-ME-12)**



22. If F = 1kN, magnitude of the vertical reaction force developed at the point B in kN is
(a) 0.63 (b) 0.32 (c) 1.26 (d) 1.46

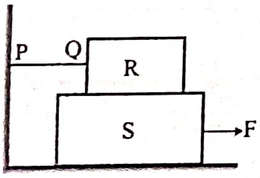
23. The maximum force F in kN that can be applied at C such that the axial stress in any of the truss members DOES NOT exceed 100MPa is
(a) 8.17 (b) 11.15 (c) 14.14 (d) 22.30

24. A pin jointed uniform rigid rod of weight W and length L is supported horizontally by an external force F as shown in the figure below. The force F is suddenly removed. At the instant of force removal, the magnitude of vertical reaction developed at the support is (GATE-ME-13)



(a) zero (b) W/4 (c) W/2 (d) W

25. A block R of mass 100 kg is placed on a block S of mass 150 kg as shown in the figure. Block R is tied to the wall by a massless and inextensible string PQ. If the coefficient of static friction for all surfaces is 0.4, the minimum force F (in kN) needed to move the block S is (GATE-14-SET-1)



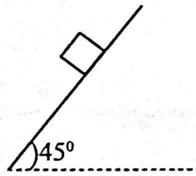
(a) 0.69 (b) 0.88 (c) 0.98 (d) 1.37

26. A block weighing 200N is in contact with a level plane whose coefficient of static and kinetic friction are 0.4 and 0.2 respectively. The block is acted upon by a horizontal force (in Newton) P = 10t, where t denotes the time in seconds. The velocity (in m/s) of the block attained after 10 seconds is ______ (GATE–14-SET-1)

27. A truck accelerates up a 10° incline with a crate of 100 kg. Value of static coefficient of friction between the crate and the truck surface is 0.3. The maximum value of acceleration (in m/s²) of the truck such that the crate does not slide down is ______ (GATE–14-SET-2)

28. A rigid link PQ of length 2 m rotates about the pinned end Q with a constant angular acceleration of 12 rad/s². When the angular velocity of the link is 4 rad/s, the magnitude of the resultant acceleration (in m/s²) of the end P is ______ (GATE–ME-14-SET-1)

29. A body of mass (m) 10 kg is initially stationary on a 45° inclined plane as shown in figure. The coefficient of dynamic friction between the body and the plane is 0.5. The body slides down the plane and attains a velocity of 20 m/s. The distance travelled (in meter) by the body along the plane is ______ (GATE-14-SET-3)

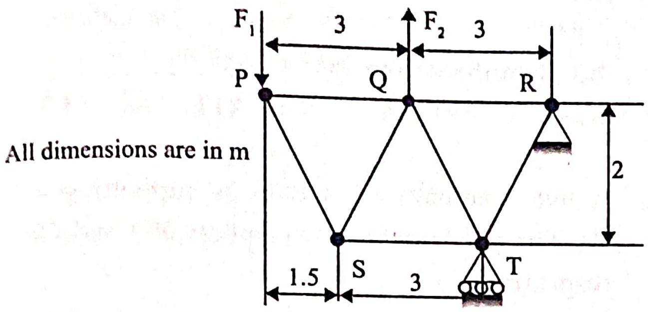


30. An annular disc has a mass m, inner radius R and outer radius 2R. The disc rolls on a flat surface without slipping. If the velocity of the center of mass is V, the kinetic energy of the disc is (GATE–14-SET-3)

(a) $\frac{9}{16}mV^2$ (b) $\frac{11}{16}mV^2$ (c) $\frac{13}{16}mV^2$ (d) $\frac{15}{16}mV^2$

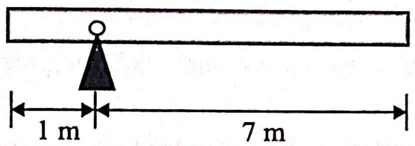
31. A four-wheel vehicle of mass 1000 kg moves uniformly in a straight line with the wheels revolving at 10 rad/s. The wheels are identical, each with a radius of 0.2 m. Then a constant braking torque is applied to all the wheels and the vehicle experiences a uniform deceleration. For the vehicle to stop in 10s, the braking torque (in N. m) on each wheel is ______. (GATE–ME-14-SET-3)

32. For the truss shown in the figure, the forces F_1 and F_2 are 9 kN and 3 kN, respectively. The force (in kN) in the member QS is **(GATE–ME-14-SET-4)**



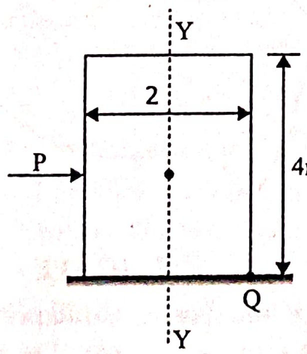
(a) 11.25 tension
(b) 11.25 compression
(c) 13.5 tension
(d) 13.5 compression

33. A uniform slender rod (8 m length and 3 kg mass) rotates in a vertical plane about a horizontal axis 1m from its end as shown in the figure. The magnitude of the angular acceleration (in rad/s²) of the rod at the position shown is _____ **(GATE–14-SET-4)**



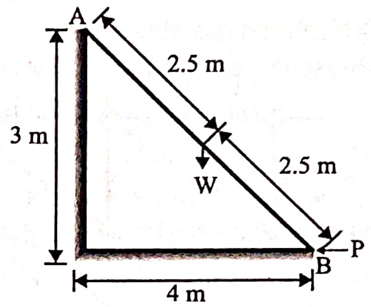
34. A wardrobe (mass 100 kg, height 4m, width 2m, depth 1m), symmetric about the Y-Y axis, stands on a rough level floor as shown in the figure. A force P is applied at mid-height on the wardrobe so as to tip it about point Q without slipping. What are the minimum values of the force (in Newton) and the static coefficient of friction μ between the floor and the wardrobe, respectively?

(GATE–ME-14-SET-4)



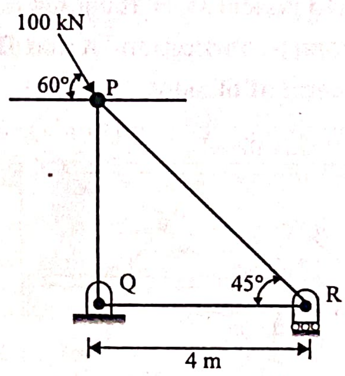
(a) 490.5 and 0.5
(b) 981 and 0.5
(c) 1000.5 and 0.15
(d) 1000.5 and 0.25

35. A ladder AB of length 5 m and weight (W) 600 N is resting against a wall. Assuming frictionless contact at the floor (B) and the wall (A), the magnitude of the force P (in Newton) required to maintain equilibrium of the ladder is _____



(GATE–14-SET-4)

36. For the truss shown in figure, the magnitude of the force in member PR and the support reaction at R are respectively **(GATE –15 –Set 1)**



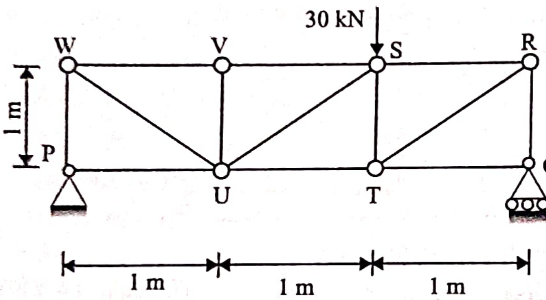
(a) 122.47 kN & 50 kN
(b) 70.71 kN & 100 kN
(c) 70.71 kN & 50 kN
(d) 81.65 kN & 100 kN

37. A ball of mass 0.1 kg, initially at rest, is dropped from height of 1m. Ball hits the ground and bounces off the ground. Upon impact with the ground, the velocity reduces by 20%. The height (in m) to which the ball will rise is _____ **(GATE –15 –Set 1)**

38. The initial velocity of an object is 40 m/s. The acceleration a of the object is given by the following expression: $a = -0.1v$. Where v is the instantaneous velocity of the object. The velocity of the object after 3 seconds will be____

(GATE –15 –Set 2)

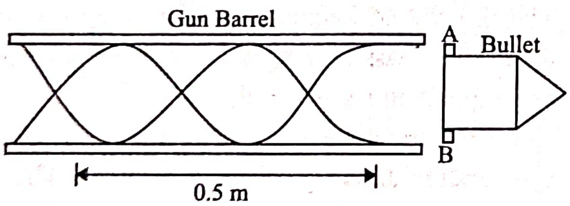
39. For the truss shown in figure, the magnitude of the force (in kN) in the member SR is



(GATE –15 –Set 2)

(a) 10 (b) 14.14 (c) 20 (d) 28.28

40. A bullet spin as the shot is fired from a gun. For this purpose, two helical slots as shown in the figure are cut in the barrel. Projections A and B on the bullet engage in each of the slots.

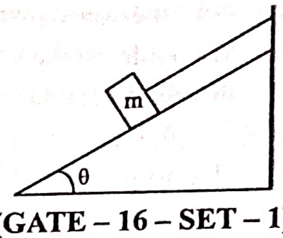


Helical slots are such that one turn of helix is completed over a distance of 0.5 m. If velocity of bullet when it exits the barrel is 20 m/s, its spinning speed (in rad/s) is_______ (GATE –15 –Set 3)

41. A block of mass m rests on an inclined plane and is attached by a string to the wall as shown in the figure. The coefficient of static friction between the plane and the block is 0.25. The string can withstand a maximum force of 20 N. The maximum value of the mass (m) for which the string will not break and the block will be in static equilibrium is ______kg.

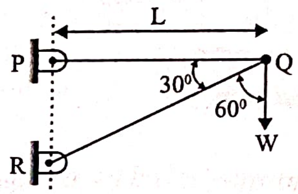
Take $\cos\theta = 0.8$ and $\sin\theta = 0.6$

Acceleration due to gravity $g = 10$ m/s²



(GATE – 16 – SET – 1)

42. A two –member truss PQR is supporting a load W. The axial forces in members PQ and QR are respectively



(GATE – 16 – SET – 1)

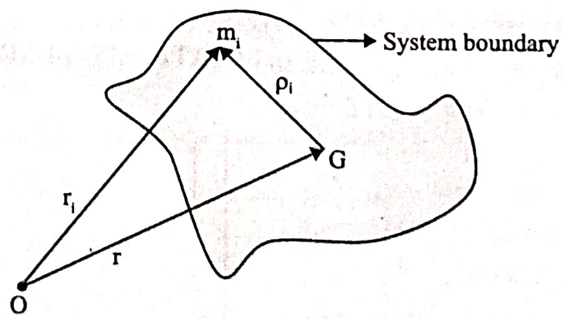
(a) 2W tensile and $\sqrt{3}$ W compressive

(b) $\sqrt{3}$ W tensile and 2W compressive

(c) $\sqrt{3}$ W compressive and 2W tensile

(d) 2W compressive and $\sqrt{3}$ W tensile

43. A system of particles in motion has mass center G as shown in the figure. The particle i has mass m_i and its position with respect to a fixed point O is given by the position vector r_i. The position of the particle with respect to G is given by the vector ρ_i. The time rate of change of the angular momentum of the system of particles about G is (The quantity $\ddot{\rho}_i$ indicates second derivative of ρ_i with respect to time and likewise for r_i).

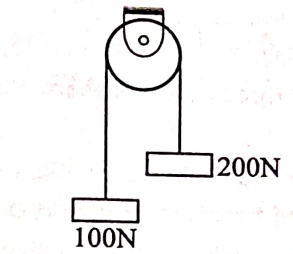


(GATE – 16 – SET – 2)

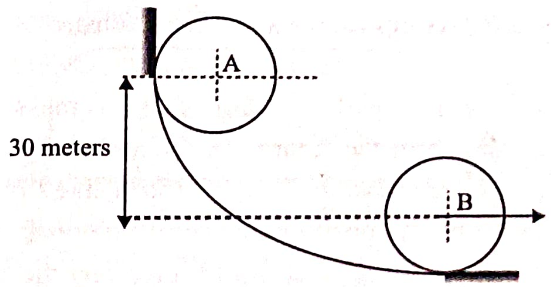
(a) $\Sigma_i r_i \times m_i \ddot{\rho}_i$ (b) $\Sigma_i \rho_i \times m_i \ddot{r}_i$

(c) $\Sigma_i r_i \times m_i \ddot{r}_i$ (d) $\Sigma_i \rho_i \times m_i \ddot{\rho}_i$

44. An inextensible massless string goes over a frictionless pulley. Two weights of 100 N and 200 N are attached to the two ends of the string. The weights are released from rest, and start moving due to gravity. The tension in the string (in N) is ________. **(GATE – 16 – SET – 3)**

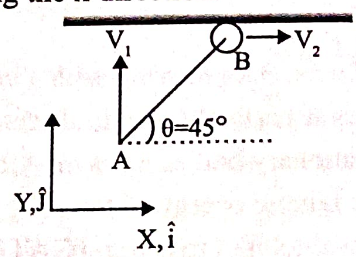


45. A circular disc of radius 100 mm and mass 1 kg, initially at rest at position A, rolls without slipping down a curved path as shown in figure. The speed v of the disc when it reaches position B is ________ m/s. **Acceleration due to gravity g = 10 m/s²**



(GATE – 16 – SET – 3)

46. A rigid rod (AB) of length $L = \sqrt{2}\,m$ is undergoing translational as well as rotational motion in the x-y plane (see the figure). The point A has the velocity $V_1 = \hat{i} + 2\hat{j}\ m/s$. The end B is constrained to move only along the x direction.



The magnitude of the velocity V_2 (in m/s) at the end B is ________ **(GATE – 16 – SET – 3)**

47. In abrasive water jet machining, the velocity of water at the exit of the orifice, before mixing with abrasives, is 800 m/s. The mass flow rate of water is 3.4 kg/min. The abrasives are added to the water jet at a rate of 0.6 kg/min with negligible velocity. Assume that at the end of the focusing tube, abrasive particles and water come out with equal velocity. Consider that there is no air in the abrasive water jet. Assuming conservation of momentum, the velocity (in m/s) of the abrasive water jet at the end of the focusing tube is ________ **(GATE - PI-16)**

48. Two disks A and B with identical mass (m) and radius (R) are initially at rest. They roll down from the top of identical inclined planes without slipping. Disk A has all of its mass concentrated at the rim, while Disk B has its mass uniformly distributed. At the bottom of the plane, the ratio of velocity of the center of disk A to the velocity of the center of disk B is **(GATE – 17 – SET – 1)**

(a) $\sqrt{\frac{3}{4}}$ (b) $\sqrt{\frac{3}{2}}$ (c) 1 (d) $\sqrt{2}$

49. The rod PQ of length $L = \sqrt{2}$ m, and uniformly distributed mass of M = 10 kg, is released from rest at the position shown in the figure. The ends slide along the frictionless faces OP and OQ. Assume acceleration due to gravity, g = 10 m/s². The mass moment of inertia of the rod about its Centre of mass and an axis perpendicular to the plane of the figure is ($ML^2/12$). At this instant, the magnitude of angular acceleration (in radian/s²) of the rod is ________ **(GATE – 17 – SET – 2)**